Looping the Loop de Loop

Skills:

All 3 loops and scanner. [/22]

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1. Determine the number of loop iterations for each of the following, and **write the answers in the top comments of your class** [K /6 marks]

   for (int i=0;i<10;i++) ___

   for (int i=1;i<10;i++) ____

   for (int i=0;i<100;i+=2) ____

   for (int i=1;i<50;i+=2) ____

   for (int i=1000;i>10;i--) ____

   for (int i=5;i<150;i+=5) ____

For the following questions, create a Java program.

2. Using a for loop, print out all the even numbers from 1 to 100. Skip Count by 1! [A /4]

3. Using a For loop print all odd numbers in descending order from 100 to 1. Skip Count by 1! [A /4]

4. Using a for loop, print a table of numbers, their squares and their cubes. Ask the user for a starting number and an ending number. [A /4]

   Sample output:

   Enter Starting number: 5
   Enter Ending Number: 10

Number	Square	Cube
5	25	125
6	36	216
.		
10	100	1000

While Loop

5. Using a while loop, play a guess the number game with the user. Have them keep guessing until they get your **secret** number. (Secret number should be randomly chosen between 1 and 100) [T /4]

Give the player clues if the number they entered is greater or smaller than the number they are trying to guess.

For example: Suppose the randomly chosen secret number is 60

Player enters 50.

Program should say the secret number is greater than 50

Player enters 70

Program should say the secret number is less than 70

Now players knows that the number > 50 and < 70 so they enter 62

Program should say the secret number is less than 62

Etc…

Until the user guess the right number

You can use the Scanner class to ask for user input

```
 //declare a new scanner call it input
     Scanner input = new Scanner(System.in);

//Prompt the user to enter their total
     System.out.println("Enter your number: ");
     //Read the amount entered
     Int number = input.nextInt();
      input.close();
```